

19. Perform Edge detection using Sobel Matrix along XY axis.

AIM:

Edge detection using Sobel Matrix along XY axis.

PROGRAM:

```
import cv2

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png', 0)

if img is None:
    print("Error: Image not found!") else:
    # Apply Sobel filter in X and Y directions    sobelx
    = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)    sobely
    = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    # Combine the Sobel X and Sobel Y images    edges
    = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

    # Convert to 8-bit image for display and saving
    edges = cv2.convertScaleAbs(edges)

    # Save the output image    cv2.imwrite(
r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Edge_detection.jpg',
edges
    )

    # Display the images
cv2.imshow("Original Image", img)
cv2.imshow("Edge Detection", edges)

cv2.waitKey(0)
cv2.destroyAllWindows() INPUT:
```



OUTPUT:



RESULT:

Thus the program is executed successfully.